Power: USB-C, charger, 3V3, rails

File: power.kicad_sch

ESP32-S3 module

File: mcu.kicad_sch

AS5311 magnetic encoder

File: sensor_mag.kicad_sch

Capacitive front-end (experimental)

File: sensor_cap.kicad_sch

Display, buttons, LED, piezo

File: ui.kicad_sch

IMU (accelerometer + gyroscope) and compass (magnetometer)

File: imu.kicad_sch

PrettyCool Caliper rev A — generated by caliper/hardware/design/build.py from caliper/hardware/design/sheets/*.py.
Do not hand-edit the .kicad_sch files: change the sheet modules and rebuild.
Architecture of record: caliper/hardware/design/SPEC.md

PrettyCool Boards

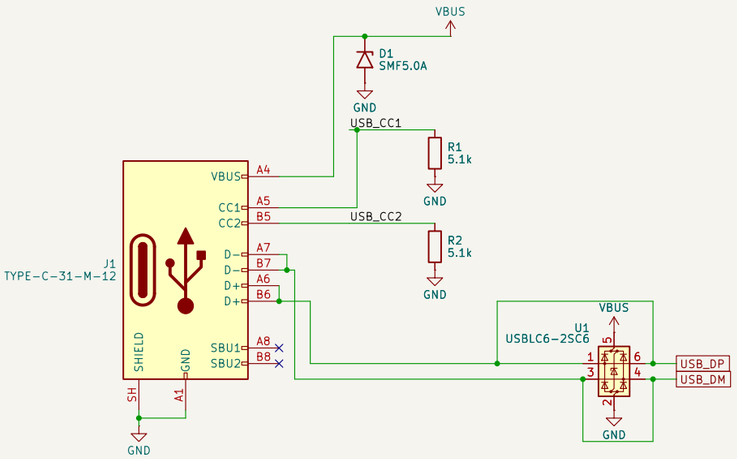
Sheet: /
File: prettycool-caliper-slider.kicad_sch

Title: PrettyCool Caliper slider board

Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/7

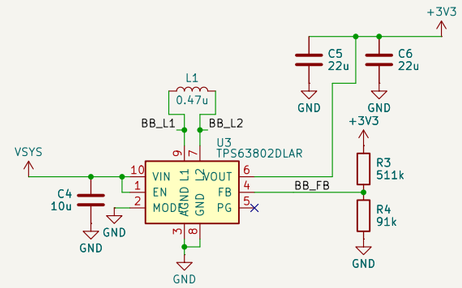
USB-C input and protection

HRO TYPE-C-31-M-12 (USB 2.0 device, sink only). Separate 5.1k Rd per CC pin; D+/D- pairs tied at the connector. USBLC6-2SC6 + SMF5.0A within 5 mm of J1, D+/D- routed through the ESD pads as a 90R pair. No series resistors on D+/D- (inside the ESP32-S3, SPEC §5). Right board edge, y 26.5 (SPEC §2.1).



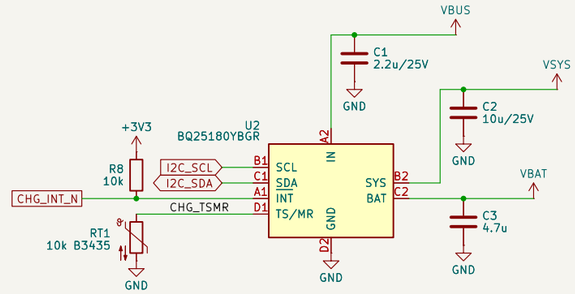
3V3 buck-boost: TPS63802 (VSYS 2.5-4.6 V -> +3V3, 2 A)

Buck-boost, not an LDO: Wi-Fi TX peaks ~355 mA (module datasheet) and the cell falls below 3.4 V (SPEC §3). EN tied to VSYS (always on), MODE = GND (power save: 11 uA Iq), PG unused. R3/R4 = 511k/91k -> 3.31 V (VFB 0.5 V, TI Table 10-5); the divider draws 5.5 uA (R2 <= 100k rule). L1 DFE201612E-R47M (TI Table 10-2). CIN 10u + COUT 22u per Fig. 10-1, plus 22u bulk C6 for the RF bursts. Layout: VIN/VOUT caps and L1 on the same layer as U3, GND pad 8 to the plane with >= 2 vias.



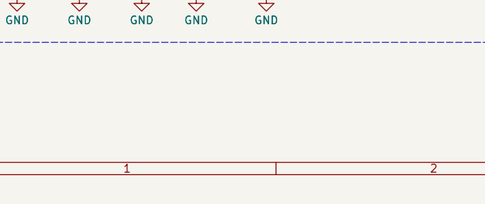
Charger and power path: BQ25180 (I2C 0x6A)

ILIM / ICHG (70 mA) / SYS_REG / ship mode set over I2C: register init per dial-me-spinner docs/firmware.md §13.1. IN 2.2u/25V X7R 0805, SYS 10u/25V, BAT 4.7u: each within 1 mm of its ball and of the D2 GND ball. Ship mode exits only on a USB plug-in (TS/MR is not wired to a button in rev A). /INT -> CHG_INT_N (10k pull-up R8, GPIO3). RT1: 10k NTC next to J2 (board temperature, not the cell).



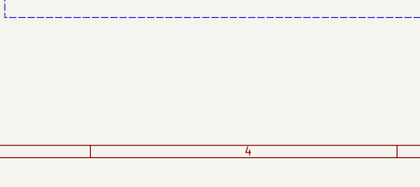
Sensor rail +3V3_SNS: TPS22917 (SNS_PWR_EN)

AS5311 (~16 mA) + capacitive front-end. Off in sleep. CT 1n to VIN (slew ~ms, no +3V3 droop): QOD tied to VOUT (internal R) so the rail discharges when off. R5 holds ON low while the ESP32 is in reset. Firmware: drive MAG_CSN / MAG_CLK low before SNS_PWR_EN goes low (no back-powering through the ESD diodes).



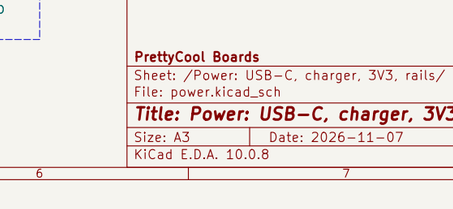
UI rail +3V3_UI: TPS22917 (also on SNS_PWR_EN)

LCD logic VDD (ST7789 sleep-in still draws 15+5 uA typ, DM-TFT114 §5.3) + WS2812 RGB LED: both off in deep sleep. Shares SNS_PWR_EN (no free GPIO): the rails are on together whenever the caliper is awake. C13 1u at VIN (pin 1): the switch's own input cap (C5 is 12.5 mm away by copper). Firmware: drive the LCD_* lines and RGB_DIN low before switching off.



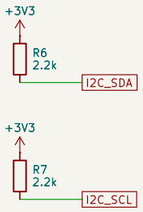
Fuel gauge: MAX17048 (I2C 0x36)

Powered from the cell (VDD = CELL = VBAT, Fig. 14). CTG, QSTRT and EP to GND. 3 uA in hibernate. ALRT -> FG_ALRT_N (10k pull-up R9, GPIO2: low SOC / voltage alert).



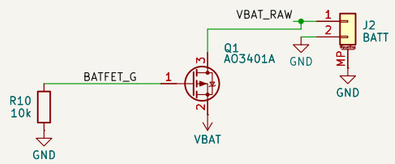
I2C + open-drain pull-ups (always-on +3V3)

SDA/SCL 2.2k: BQ25180 0x6A, MAX17048 0x36, LSM6DSO 0x6B at 400 kHz (SPEC §6): ~205 ns rise (30-70 %) with ~110 pF (4.7k gave 438 ns > 300 ns); 1.3 mA sink at VOL 0.4 V (fast mode allows 3 mA). Must stay on the always-on rail: the MAX17048 reads a low bus as 'disconnected' and sleeps.



LiPo connector J2 (~135 mAh 1S, JST-SH 2-pin) + reverse-polarity P-FET Q1

Pin 1 = BAT+, pin 2 = GND. Pack leads are not standardised: check every pack's polarity anyway. Q1 AO3401A in the BAT+ lead: drain to J2, source to VBAT, gate to GND via R10. The body diode conducts for a correct cell, then VGS = -VBAT turns the channel on (both directions, so charging works). A reversed cell leaves VGS = 0: off. VGS max +-12 V vs 4.4 V: no zener. 0.5 A: <= 43 mV / 21 mW (85 mOhm max at VGS -2.5 V). J2 is on the bottom: it, the plug and the cell sit in the housing pocket behind the board (SPEC §2.1).



Mounting holes and fiducials

H1 (18.75, 24) support hole; H2-H5 tab clamp holes (-3.5, 5.5) (-3.5, 14) (65.5, 5.5) (65.5, 14), plated, to GND (SPEC §2.1). H2-H5 carry an SMT M2 standoff, fitted by hand (no LCSC part fits the 6 mm tab). 3 fiducials.



Power flags and test points

VBUS (USB-C) and GND are driven from outside the board.



PrettyCool Boards

Sheet: /Power: USB-C, charger, 3V3, rails/
File: power.kicad_sch

Title: Power: USB-C, charger, 3V3, rails

Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 2/7

ESP32-S3-MINI-1-N4R2 (4 MB flash, 2 MB PSRAM, PCB antenna)

Pin map = nets.MCU_PINS (SPEC §5); caliper/hardware/design/check_contract.py checks the netlist against it.
Antenna end on the top board edge (y = 34), hanging clear of the steel beam; copper keep-out on all layers under the antenna (Espressif HW design guidelines). EPAD + GND pads to the plane with a via array.
N8R2 is not stocked at LCSC (2026-11); N4R2 C3013941 is. IO26 is the PSRAM CS: left unconnected.

3V3 decoupling at the module

Datasheet Fig. 9-1: 22u + 100n at pad 3. C21 100n + C23 10u (0402, 0.5 mm tall) on the BOTTOM, under pad 3, fed from its escape via, own GND vias (the housing floor is 0.75 mm below there, SPEC §2.1).
C20 22u (0603, 0.8 mm) is the bulk on top below the module, fed through the In2 +3V3 pour.

EN reset delay

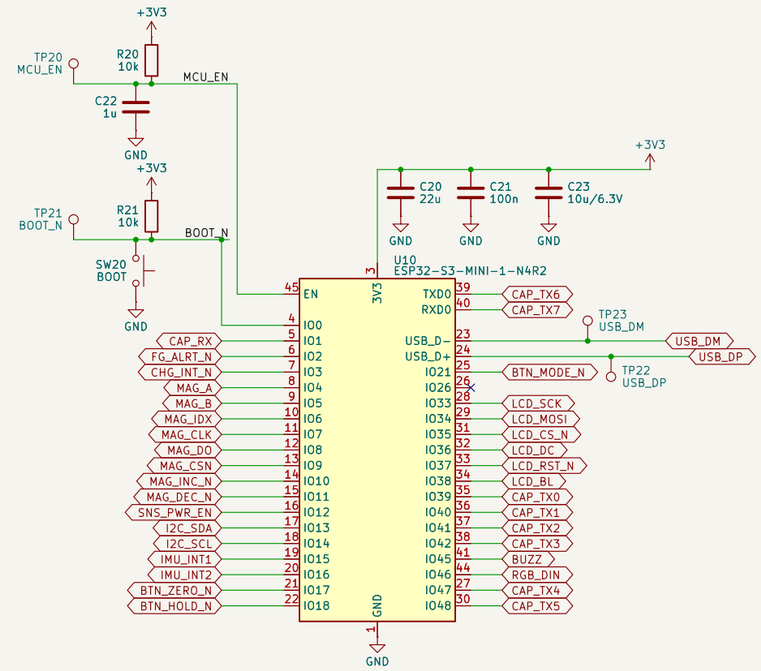
R 10k / C 1u (datasheet §9). EN pad on a test point for a manual reset.

BOOT (GPIO0 strap)

10k pull-up: SPI boot. Hold SW20 during reset/USB plug-in for the ROM download mode.
Normally flashed over USB-Serial-JTAG without the button; SW20 is the recovery path.

Test points

EN, BOOT and the native USB pair for bring-up



Strapping pull-downs

GPIO45 = VDD_SPI select, must read 0 at reset (3.3 V flash): 100k on the BUZZ gate is on the ui sheet (R56).
GPIO46 must be 0/float at reset: 100k on RGB_DIN is on the ui sheet (R55).
GPIO3 (CHG_INT_N, 10k pull-up R8): JTAG-source strap, ignored while the JTAG_SEL eFuse is unburnt.

PrettyCool Boards

Sheet: /ESP32-S3 module/
File: mcu.kicad_sch

Title: ESP32-S3 module

Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 3/7

AS5311 linear encoder (BOTTOM side, over the magnetic tape)

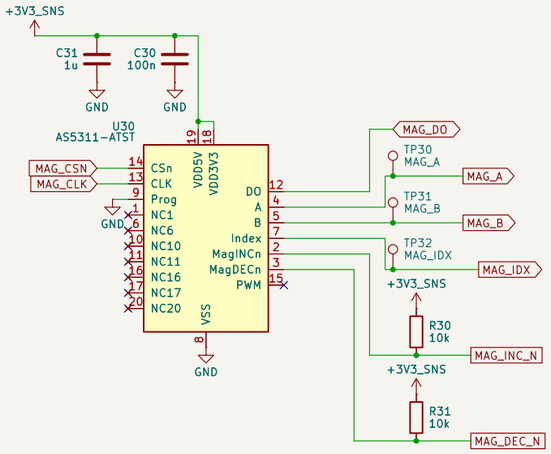
Centred at (31, 6.0) on the bottom, long axis along X (beam axis), SPEC §2.1. Air gap 0.3 +/−0.1 mm to the tape.
3.3 V operation: VDD5V and VDD3V3 both on +3V3_SNS (datasheet §7.7), 100n + 1u at pin 19.
A/B −> PCNT (4x: 0.488 um/count), Index once per 2 mm pole pair, SSI (CSn/CLK/DO) for the 12−bit absolute position + OCF/COF/LIN/MagINC/MagDEC. CSn must be low for A/B/Index. Prog to VSS (no OTP in rev A). PWM unused.
No copper pour, vias or magnetic parts in the tape window under the IC beyond what the bottom layer rules allow.

Magnetic−field indicator pull−ups

MagINCn / MagDECn are open drain (datasheet Table 1, §4.1): 10k to +3V3_SNS, so they read low (= 'alarm') when the sensor rail is off; the firmware only reads them with SNS_PWR_EN high.

Test points

Quadrature + index for a scope / logic analyser during bring−up



PrettyCool Boards

Sheet: /AS5311 magnetic encoder/
File: sensor_mag.kicad_sch

Title: AS5311 magnetic encoder

Size: A3	Date: 2026−11−07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 4/7

TX phase isolation jumpers (0R, DNP to disconnect the experiment)

Fit R60–R67 to drive the electrodes; leave them off and the eight GPIOs are free test pads.

TX series resistors (100R: edge-rate limit, EMI, GPIO protection)

Next to the comb, on the electrode side of the jumpers. Phase n drives every 8th finger (pitch 0.635 mm).

Electrodes (bottom copper, E40 = seed pads; not a BOM part)

Comb x = 11…51, y = 12.5…17.5; RX strip y = 10.6…11.8 (SPEC §2.1). Bottom mask open over both.
Guard GND around the RX strip and the CAP_EL_RX trace on all layers; nothing else on the bottom in y = 9…19.

RX jumper + charge amplifier (U40A)

*Q = C*V: Cf 10p C0G || Rf 10M (HPF corner 1.6 kHz, far below the 25–100 kHz drive). Output centred on VREF.*
Keep the CAP_RX_IN node tiny, guarded, no mask opening on it except at the electrode.

VREF = +3V3_SNS / 2, buffered (U40B)

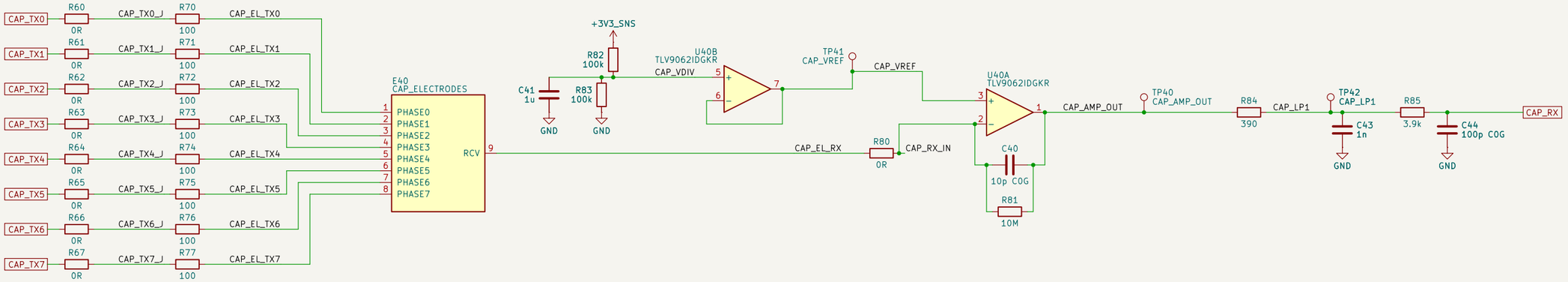
100k/100k divider (16.5 uA, only while +3V3_SNS is on) filtered by C41 1u, unity-gain buffer U40B.
The buffer output drives only U40A's + input (no capacitive load on the op-amp output).

2nd-order RC low-pass (~408 kHz per pole) -> CAP_RX (ADC1_CH0)

390R/1n then 3.9k/100p C0G (second stage 10x the impedance, so it barely loads the first).
Anti-alias + kills the op-amp's HF noise; the synchronous demodulation is done in firmware.
Layout: R84/C43 at U40, R85/C44 at the ESP32's GPIO1 pad (bottom, under the module): the long run
is CAP_LP1 (390R + 1n at the source end) and CAP_RX is a few mm from the ADC pin.

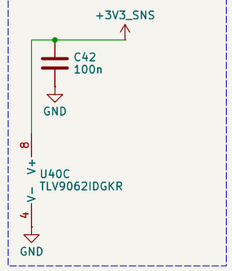
Test points

Charge-amp output, VREF and the filter's first stage (CAP_LP1) for scope probing.
TP42 is on CAP_LP1, not CAP_RX: CAP_RX ends at the ESP32 (R85/C44 there), and a pad in the
test-pad row would hang a 30 mm stub on the ADC node.



Op-amp supply

100n at pin 8



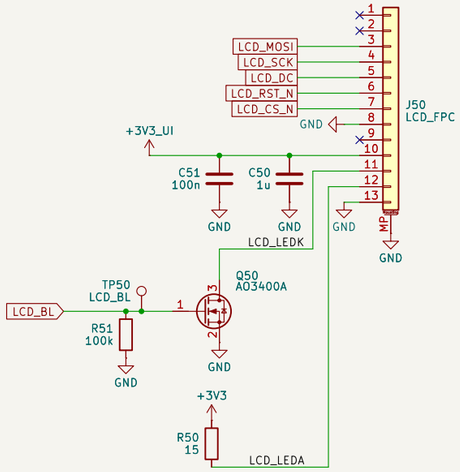
PrettyCool Boards		
Sheet: /Capacitive front-end (experimental)/		
File: sensor_cap.kicad_sch		
Title: Capacitive front-end (experimental)		
Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 5/7

LCD: 1.14in 240x135 ST7789 IPS, 13-pin 0.5 mm FPC (J50)

Pin map: DM-TFT114-397 §3.1 (1 NC, 2 NC, 3 SDA, 4 SCL, 5 RS=D/C, 6 RESET, 7 CS, 8 GND, 9 NC, 10 VDD, 11 LEDK, 12 LEDA, 13 GND).
SPI2 40 MHz write-only (SPEC §6). VDD from the switched +3V3_UI rail. 1u + 100n at pin 10.
Beside the module on top; the glass is glued to the bezel, not the board (SPEC §2.1).
Check the bought panel's pin order and 0.5 mm tail pitch before ordering (vendors differ).

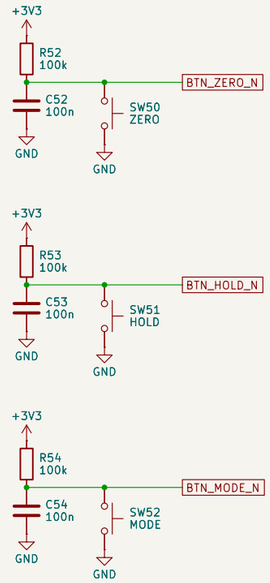
Backlight: low-side N-MOSFET on LCD_BL (PWM)

+3V3 -> R50 15R -> LEDA, LEDK -> Q50 -> GND. 1 white LED, VF 3.0 V (2.9-3.1) at 20 mA.
(3.31 V - VF) / 15R: 20 mA at VF 3.0, 27 mA max at 2.9, 14 mA at 3.1 (10R gave 31-41 mA).
Headroom is only ~0.3 V, so brightness follows VF and the +3V3 tolerance; dim by PWM (LEDC, ~20 kHz).
R51 100k holds the gate low while GPIO38 floats in reset.



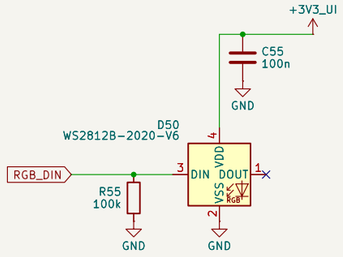
Buttons: ZERO / HOLD-SEND / MODE (active low, RTC GPIOs 17/18/21 for deep-sleep wake)

100k pull-up (no static current) + 100n to GND: 10 ms RC debounce; ext1 wake on any-low.
TS-1088 3.9x2.9 mm top tact switches on the top edge band: ZERO x-8, HOLD x-20, MODE x-54 (SPEC §2.1).



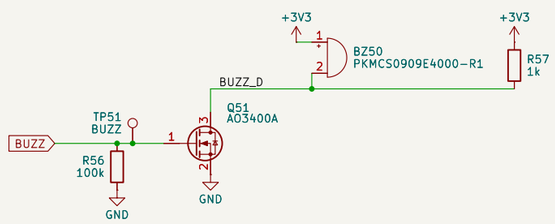
RGB status LED: WS2812B-2020-V6 on +3V3_UI (GPIO46 = RGB_DIN)

3.3-5.3 V part (LCSC C52917434), so no level shifter: DIN at 3.3 V logic = VDD. Off with +3V3_UI in sleep.
R55 100k pull-down: GPIO46 is a strap (must be 0/float at reset). 100n at VDD.



Piezo: Murata PKMCS0909E4000-R1 (4 kHz, external drive), low-side MOSFET on BUZZ (GPIO45)

+3V3 -> BZ50 || R57 1k -> Q51 drain. LEDC square wave at ~4 kHz on the gate. R57 recharges the piezo
while Q51 is off (3.3 mA only while Q51 conducts). R56 100k gate pull-down: GPIO45 is the VDD_SPI strap
and must read 0 at reset (SPEC §5).



PrettyCool Boards

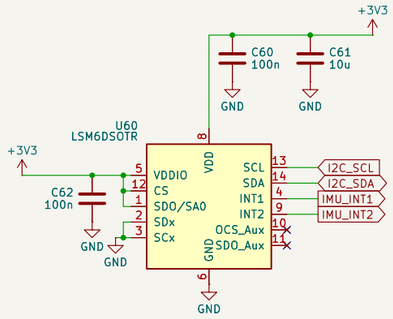
Sheet: /Display, buttons, LED, piezo/
File: ui.kicad_sch

Title: Display, buttons, LED, piezo

Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 6/7

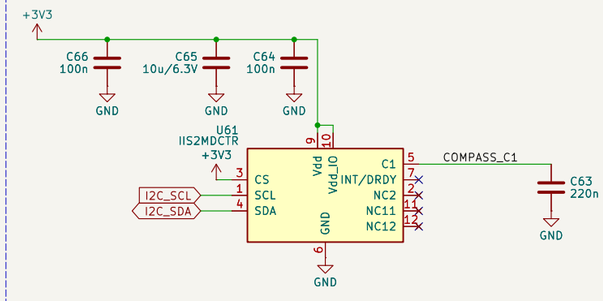
6-axis IMU: LSM6DSO on the system I2C (addr 0x6B, SA0 high; 0x6A is the BQ25180)

Always on (+3V3). Asleep: accelerometer only, ultra-low-power 12.5 Hz, gyroscope powered down (5.5 uA typ, AN5192 Table 9), wake-up interrupt while the ESP32 is in deep sleep. Awake: gyroscope on (0.30 mA at 52 Hz low-power). CS high = I2C. SA0 high (no internal pull-up by default). SDx / SCx to GND, OCS_Aux / SDO_Aux unconnected. INT1 -> IMU_INT1 (GPIO15, RTC: deep-sleep wake on pickup), INT2 -> IMU_INT2 (GPIO16). Push-pull, no pulls: INT1 must be low or floating while the IMU powers up (its own pull-down holds it). 100n + 10u at VDD (pin 8), 100n at VDDIO (pin 5). Keep away from the buck-boost inductor.



Compass: IIS2MDC 3-axis magnetometer on the system I2C (addr 0x1E, fixed)

Always on (+3V3). Polled over I2C: INT/DRDY is not connected (no free ESP32 pin; high impedance by default). Awake: low-power mode, 10 Hz, offset cancellation on (50 uA, datasheet Table 11). Asleep: idle (1.5 uA typ). CS high = I2C. C1: 220n X7R right at pins 5 / 6 (current pulses). 100n + 10u at Vdd (pin 9), 100n at Vdd_IO (pin 10). Placed in the top band, as far from the steel beam and the magnetic tape as the board allows (SPEC 17). Keep tracks that carry more than 10 mA a few millimetres away (datasheet 5.2).



PrettyCool Boards

Sheet: /IMU (accelerometer + gyroscope) and compass (magnetometer)/
File: imu.kicad_sch

Title: IMU (accelerometer + gyroscope) and compass (magnetometer)

Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 7/7